Atharva Kirkole

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EDUCATION

Michigan State University East Lansing, MI
M.S. Computer Science — GPA 3.93 | TA (Software Performance Engineering, OOP) *Aug 2025 – May 2026*

Michigan State University East Lansing, MI
B.S. Computer Science, Minor in Business (Honors College) — GPA 3.93, Honors, Tau Beta Pi *Aug 2022 – Apr 2025*

EXPERIENCE

Michigan State University East Lansing, MI
Machine Learning Researcher — Speech AI, Scientific ML & HPC *Feb 2026 – Present*

- **Neural mobility operator (NeMO)** — learned surrogate for the dense operator mapping forces and torques to velocities for N rigid particles in Stokes flow, replacing an $O(N^2)$–$O(N^3)$ boundary-element solve; analytic far field plus PyTorch/CUDA-learned near-field lubrication and many-body corrections, Cut velocity error from **12.8% → 2.5%.**
- **Extended the learned operator's output basis** from 5 rotation-invariant coefficients on a single pooled direction to a 17-term O(3)-equivariant tensor basis on a DeepSet encoder.
- Co-developed the **warp-specialized producer–consumer CUDA model** (one traversal warp, 15 consumer warps for M2P/P2P off a lock-free shared-memory ring), minimizing thread idling on irregular workloads; **cuts 32 serialised atomics to one per warp, sustaining 87.7% lane utilization.** Parallelized the upward pass bottom-up via LBVH, eliminating per-level sync.
- ***Speech AI (Heard AI / NSF):*** contributed to a **$5M NSF Convergence Accelerator** (Phase 2) project on accessible speech technologies; built speaker diarization and stutter-detection models to segment multi-speaker audio and classify dysfluent speech patterns (repetitions, prolongations, blocks) from raw waveforms.
- Engineered a stutter speech synthesis/regeneration pipeline converting dysfluent input into fluent alternatives and shipped the full-stack application end to end (**React, Django, Flask**) sustaining **300 concurrent sessions at 400 ms** latency via async inference queues and streaming audio transport.
- Implemented a per-user fine-tuning framework using rapid low-data speaker adaptation, cutting word error rate **23% (14% → 10.8%)** while keeping per-user adapter footprints small enough for on-demand loading at inference time.

Michigan State University IT Services East Lansing, MI
Software Developer *Jun 2024 – Apr 2026*

- Led end-to-end migration of the campus mapping platform off Google Maps onto a self-managed ESRI ArcGIS stack serving **100K+ daily requests**, rebuilding map surface around custom geospatial layers and cutting licensing cost ~20% (~**$45K/yr**).
- Owned geospatial data modeling for ~1.2k+ campus entities/POIs
- Engineered student/people information-system APIs and Dockerized service images (PHP, JavaScript); raised site accessibility from 80% to 99.7% (WCAG 2.1 / ADA / ARIA) across 40 production surfaces.
- Built an interactive browser-based malaria transmission simulator in modular ES6 JavaScript — vector–host compartment model with client-side numerical integration, parameter controls, and live plotting.

Amazon East Lansing, MI
University Capstone Project *Jan 2025 – May 2025*

- Built an AI-powered semantic code-search system over **~10k** internal architecture assets on AWS OpenSearch (Elasticsearch-compatible), Lambda, DynamoDB, and S3, exposed through an **API Gateway** + Lambda REST layer with **FastAPI and Flask** inference services and a **React** search UI backed by **Node.js**.
- Designed a RAG-style retrieval pipeline (dynamically chunked vector embeddings via Bedrock + SageMaker, indexed in OpenSearch); tuned indexing and ranking to lift top-k relevance **~18%,** added vector-similarity employee recommendations over GitHub signals, and containerized the **FastAPI** services with Docker for reproducible deployment.

PROJECTS

Intra building Navigator — MSU AI Club- LINK 2022

- **Built an indoor navigation system** providing turn-by-turn guidance inside building interiors, where GPS fails — Flask/Python backend serving a Flutter cross-platform client over HTTP, with floor-plan parsing, location resolution and pathfinding.
- **Implemented image-based positioning** using EXIF metadata extraction and GeoPy geolocation to resolve a user's position from a captured photo, removing the need for beacon hardware or pre-mapped WiFi fingerprints.

OceanGPU — GPU-Accelerated Ocean Wave Simulation — LINK **2026**

- Built an open-source, zero-dependency **WebGPU/WGSL** library solving a real-time 2D wave-equation PDE with a **shared-memory tiled compute kernel** (16×16 workgroups, 18×18 halo tile, workgroupBarrier); **verified 60 fps at 368,640 GPU threads/frame**, WebGL2 fallback for full browser coverage.
- Shipped it as a **one-call embeddable API + interactive playground** (10 themes, live controls, embed-code generator) with a stress lab scaling to **16.8M cells / 512 sources**; deployed on GitHub Pages, headless-Chrome validated.

Distributed Systems(MPI) – Masters Parallel Computing Course (4.0) 2026

- Latency hiding: overlapped non-blocking communication (Isend/Irecv) with stencil computation
- One-sided RMA: MPI_Win/MPI_Get for direct array access; performance gains on large datasets (100M+ elements/rank)
- Custom data types: MPI_Type_create_struct with offsetof for efficient MPI_Gather & Subcommunicators: MPI_Comm_split

TECHNICAL SKILLS

Maps & Geospatial: ESRI ArcGIS, ArcGIS Portal, Google Maps API, Leaflet.js
Tools: Hadoop, AWS, OpenSearch, Bedrock, EC2, RAG, DynamoDB, MongoDB, S3
Languages & ML/Web: C++17, C#, Python, JavaScript, SQL, R, PHP; PyTorch, TensorFlow, React, Node.js, Flask, Django, REST APIs, Docker, AWS Lambda; CUDA / OpenMP / MPI, PostgreSQL, Molecular Dynamics